

BLOEMFONTEIN CAMPUS

CSIS3754

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

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SEMESTER TEST 2

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MODERATOR(S): 1. Prof. J.V. du Toit

TIME: 180 min

MARKS: 90

INSTRUCTIONS:

- This question paper has two questions:
 - Answer Question 1 on the answer sheet provided. (Closed Book)
 - Create and submit a Jupyter Notebook for Question 2. (Open Book)
- Submit the notebook to the following Blackboard link:
 - CSIS3754 -> Assessments -> Semester Test 2 -> Question 2.

Question 1 [20 Marks - 40 Minutes] (Closed Book)

1.1 Consider the following learning curves of two machine learning models. On the left, a learning curve of a Support Vector Machine classifier and on the right, a learning curve of a K-Nearest Neighbour classifier:



1.1.1 Which model would most probably provide better prediction results? (1)

1.1.2 Motivate your answer in 1.1.1 by explaining why the relevant model would provide better prediction results. (3)

- 1.2 Explain the steps that are involved when the k-Means clustering algorithm is used to group data points into clusters. (5)
- 1.3 Consider the data table below that can be used to train a decision tree capable of determining whether a person is likely to suffer a heart attack depending on the person's blood pressure and age:

High blood pressure (HBP)	Younger than 48 (Age < 48)	Heart attack (HA)
Yes	Yes	Yes
No	Yes	No
Yes	Yes	No
No	Yes	No
Yes	No	Yes
Yes	Yes	No
Yes	No	Yes

- 1.3.1 Using the Gini Impurity metric, determine the root feature of the decision tree. Show all your calculations. (11)
- [20]

Question 2 [70 Marks - 140 Minutes] (Open Book)

Please log into Blackboard and go to CSIS3754 -> Assessments -> Semester Test 2 -> Question 2 and download the data file named *credit_card_customers.csv*. The data contains information about bank customers who own credit cards. Create a Jupyter notebook that contains the Python code to construct machine learning classifiers that are able to determine whether a current customer will stay with the bank or intends to leave, depending on the feature values for each specific customer. Name the notebook "CSIS3754_YourStudentNumber_ST2-2_2025".

The Jupyter notebook should have the following functionality (NB!!! - Show the changes in the data after each operation):

- 2.1 Import the data file into a dataframe named "customers" with the column headings as contained in the data file. (1)
- 2.2 Inspect the data by: (4)
- Determining and displaying the row count and column count of the dataset.
 - Displaying the last 20 records of the dataset.
 - Displaying the data type for each column.
 - Displaying the number of unique values for each feature.
- 2.3 Discard the first, as well as last 7 columns of the dataset without explicitly specifying the column names. (2)
- 2.4 According to your analysis of the dataset, what does the feature *Avg_Open_To_Buy* refer to? (Print your answer in the Jupyter notebook) (2)

- 2.5 Generate a statistical summary of all the features. (1)
- 2.5.1 What can be deduced from the statistical summary with regard to the customers' education level? (Print your answer in the Jupyter notebook) (2)
- 2.6 Determine and display the percentage of customers who receive an income of R120 000 (R120K) or more for: (4)
- Uneducated customers and
 - Customers with a Doctor's (PhD) degree.
- 2.6.1 What conclusion can be made from the results obtained in 2.6? (Print your answer in the Jupyter notebook) (1)
- 2.7 Visualise the data by using the seaborn library to create the following plots. Below each plot, discuss the information that can be deduced from the relevant plot: (Print your answer in the Jupyter notebook) (9)
- Create a category plot that shows the number of customers belonging to a specific Card Category for each Education Level.
 - Create a scatter plot between the customers' Credit Limit and the *Avg_Open_To_Buy* column values.
 - Create a single combined scatter plot between the customers' Income Category and their Credit Limit for all the different Card Categories.
- 2.8 The *Attrition_Flag* column value indicates whether the relevant record belongs to an existing customer or to someone who is no longer a customer (attrited customer). Determine and display the number of records with regard to existing customers and attrited customers. (1)
- 2.9 Determine whether the data is balanced with regard to the number of existing customers and attrited customers by: (4)
- Creating a pie chart that shows the percentage of records for each customer group.
 - Percentages displayed on the pie chart should be rounded to 3 decimals.
- 2.10 Perform the necessary operations to balance the data: (5)
- Use an undersampling technique to balance the number of records with regard to existing customers and attrited customers.
 - Assign the resampled data to the dataframe named "*customers_resampled*".
- 2.11 Confirm the successful resampling of the data by: (3)
- Displaying the number of records with regard to existing customers and attrited customers.
 - Recreating a pie chart that shows the updated percentage of records for each customer group.
 - Percentages displayed on the pie chart should be rounded to 3 decimals.
- 2.12 Convert the text values in the dataframe to numeric values: (7)
- Convert the columns that contain only two unique text values to binary values [0, 1].
 - Convert the columns that contain more than two unique text values to a vector space using one-hot encoding.

- 2.13 Define X and y and create a training / testing dataset of 80% / 20%: (7)
- X = Data features.
 - y = Attrition_Flag.
 - Determine the dimensions of X_train and X_test.
 - Ensure that all pre-processing is complete.
- 2.14 Train the following classifiers and determine the best model by using k-fold cross-validation: (10)
- K nearest neighbour (default values).
 - Logistic Regression (default values).
 - Support Vector Machine (gamma = auto).
 - Set k = 5 for the k-fold cross-validation.
 - Report the cross-validated training accuracy and F1 scores for all the classifiers.
 - Draw a learning curve for each of the classifiers.
- 2.15 Select the model that produced the highest F1 score to do the following: (3)
- Make predictions using the test dataset.
 - Provide the test accuracy score, confusion matrix and classification report of the model.
- 2.16 Discuss the selected model's metrics and scores with regard to the following: (4)
(Print your answer in the Jupyter notebook)
- Discuss the model's training accuracy in comparison to its testing accuracy.
 - Discuss the meaning of the precision, recall and F1-scores.
- [70]

End of Paper